

Continuous functions – 7

Compactness, extrema and semicontinuity

Homework

1 сентября 2026 года

Theoretical minimum. Compactness in $\mathbb{R}^n$: Heine–Borel, Bolzano–Weierstrass and the finite subcover definition; Weierstrass’ theorem and its semicontinuous refinement; uniform continuity on compacts; coercivity; Dini’s theorem; the distance function to a set and its Lipschitz property; nested compact sets.

1. Prove that the three definitions of compactness in $\mathbb{R}^n$ agree, and give an example of a closed bounded set in $C[0, 1]$ which is not compact.
2. Prove that $\text{dist}(x, A) = \inf_{a \in A} |x - a|$ is 1-Lipschitz for every non-empty A , that the infimum is attained when A is closed, and that the point realising it is unique when A is convex and closed.
3. Let $K_1 \supseteq K_2 \supseteq \dots$ be non-empty nested compact sets. Prove that $\bigcap K_n \neq \emptyset$, and give an example of nested non-empty closed sets with empty intersection.
4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous and suppose that f maps every closed set to a closed set. Prove that the preimage of every compact set is compact, provided f is also injective.
5. Prove that a lower semicontinuous function is the pointwise supremum of a family of continuous functions, and deduce that a pointwise supremum of continuous functions is lower semicontinuous.
6. Let $f: [0, 1] \rightarrow \mathbb{R}$ be such that every level set $\{f = c\}$ is closed. Must f be continuous? Prove it or give a counterexample.
7. Let f be continuous on a compact K and injective. Prove that $f^{-1}: f(K) \rightarrow K$ is continuous, and give an example on a non-compact set where the inverse fails to be continuous.

8. Prove that a continuous function on a compact metric space is uniformly continuous, and deduce that for every $\varepsilon > 0$ a continuous function on $[0, 1]$ can be approximated within ε by a piecewise-linear function.
9. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous with $\lim_{|x| \rightarrow \infty} \frac{f(x)}{|x|} = +\infty$. Prove that for every y the function $x \mapsto f(x) - \langle y, x \rangle$ attains its minimum, and explain the connection with the Legendre transform.
10. Let $K \subseteq \mathbb{R}^n$ be compact and let $f: K \rightarrow K$ satisfy $|f(x) - f(y)| < |x - y|$ for all $x \neq y$. Prove that f has a unique fixed point and that the iterates converge to it.